Where Bravo's money actually goes

> *What does Bravo cost on a like-for-like basis with LORA? What drives it? Which levers are real, how much are they worth, and when are they available?*

Rp 58M THE LOS ORCHESTRATION TIER — AGAINST LORA'S RP 431M	**Rp 140.5M** CLOUD LOGGING — 2.4× THE WHOLE ORCHESTRATION TIER	**Rp 573.5M** NON-PRODUCTION — 31% OF ALL BRAVO SPEND, NEVER EXAMINED	**4.5%** SHARE OF ITS OWN PROJECT THAT THE WORKFLOW ENGINE REPRESENTS

Audience Platform, finance, the CTO office sizing the migration **Sources** GCP billing via `bfi-finops-dashboard` · complete month of August 2026 **Read** 2026-09-10

Six claims, graded against an independent billing pull

Claim	Verdict	Evidence
Bravo is expensive	● NOT AS AN ORCHESTRATION TIER	**It is the cheaper one.** `ms-bpm` pods plus its Cloud SQL instance is **≈Rp58M/month in production** against LORA's ≈**Rp431M all-in** — roughly **7× cheaper** in absolute terms and **4–5× cheaper per application**. This document does not overturn that; it confirms it with an independent pull.
Retiring Bravo saves ≈Rp1.6B/month	● WITHDRAWN	And re-confirmed as withdrawn. The Bravo GCP projects bill **Rp1.874B** in August, but the remainder beyond the LOS tier is the **shared BFI data plane** — ~26 `ms-*` services that *LORA itself calls*. The LOS lever is **≈Rp58M prod, ≈Rp70–100M with non-prod**: 8–14× the entire Temporal Actions programme, not 227×.
Bravo's cost is dominated by the workflow engine	● NO	The engine tier is **4.5% of its own production project**. Cloud SQL (Rp450.7M) and Compute Engine (Rp394.4M) are 65% of `bravo-project-331802`, and neither is mostly BPM.
Bravo's unit economics are improving as it drains	● REFUTED	Volume fell **35%** (117,996 → 76,446 applications) between July and August while ticket load **rose 82%** over eight months. Fixed platform cost over falling volume means cost per application is **inflating**, for reasons unrelated to architecture.
The cost work worth doing is on the engine	● REFUTED — THREE LARGER LEVERS	**Cloud Logging** at **Rp140.5M/month** — 2.4× the whole orchestration tier — sits behind a globally enabled `loggerLevel: full` and an error stream that is 47% stack-trace frames. The Maps API stack is **Rp64.0M/month**. Non-production is **Rp573.5M**.
Bravo's cost is a reason to keep it	● PARTLY — AND THE STRONGEST ARGUMENT ON ITS SIDE	But the tier that would retire is small (≈Rp58M) and the platform it would retire *to* costs more per application today. **The cost case for migration is that LORA's marginal cost is near zero and one platform goes away** — not that Bravo is expensive.

> ## Bravo's LOS orchestration is genuinely cheap, and almost none of Bravo's bill is orchestration.
>
> The money is in a database, a log pipeline and a non-production estate — and two of those three are misconfiguration rather than workload.

Bravo has no metered orchestration invoice

There is no Actions counter, no per-workflow storage line, nothing analogous to Temporal Cloud. Camunda 7 is an **embedded library** inside a Spring Boot service, so every cost it creates lands somewhere else on the GCP bill.

WHERE THE ENGINE'S COST SHOWS UP	LINE	NOTES
Process state, history, and all 238 business entities	**Cloud SQL** — `prod-postgres-bpm-d2bpm`	History level `full`, `historyTimeToLive: P90D`. `act_hi_actinst` carries ~50M rows per 90-day window
Job-executor threads running all 483 service tasks	**Compute Engine** — `ms-bpm` pods on `prod-bravo-cluster`	Job executor left at the Spring Boot starter default — core pool 3
Every log line the engine and its controllers emit	**Cloud Logging**	1,116,135 lines/week from `prod-ms-bpm` alone
Outbound calls to 25 upstreams	those services' own bills	—

THE COMPARISON THIS MAKES POSSIBLE — AND THE TRAP IN IT

Because the engine is embedded, "the cost of Bravo's orchestration" is not separable from "the cost of Bravo's human-task application": **they are the same deployment.** `ms-bpm` contains the BPMN engine *and* the ~217k LOC of surveyor, operation, underwriting and approval services — 44% of the codebase. So the ≈Rp58M tier is **orchestration plus all human-task handling**, which is the correct like-for-like against LORA's ≈Rp431M (workers + gateway + task service + schema service + Temporal + ArangoDB). **Both sides include the human-task layer. That is what makes the ratio meaningful.**

August 2026, measured

PROJECT	AUGUST 2026
`bravo-project-331802` — production	**Rp 1,300,270,632**
`bravo-project-nonprod`	**Rp 573,504,402**
`bravo-project-shared-services`	Rp 13,568,891
`bravo-project-collection`	Rp 181,361
Bravo total	**Rp 1,887,525,286**
BFI GCP estate, all 34 projects	*Rp 3,708,544,980*

BRAVO IS 50.9% OF BFI'S ENTIRE GCP BILL

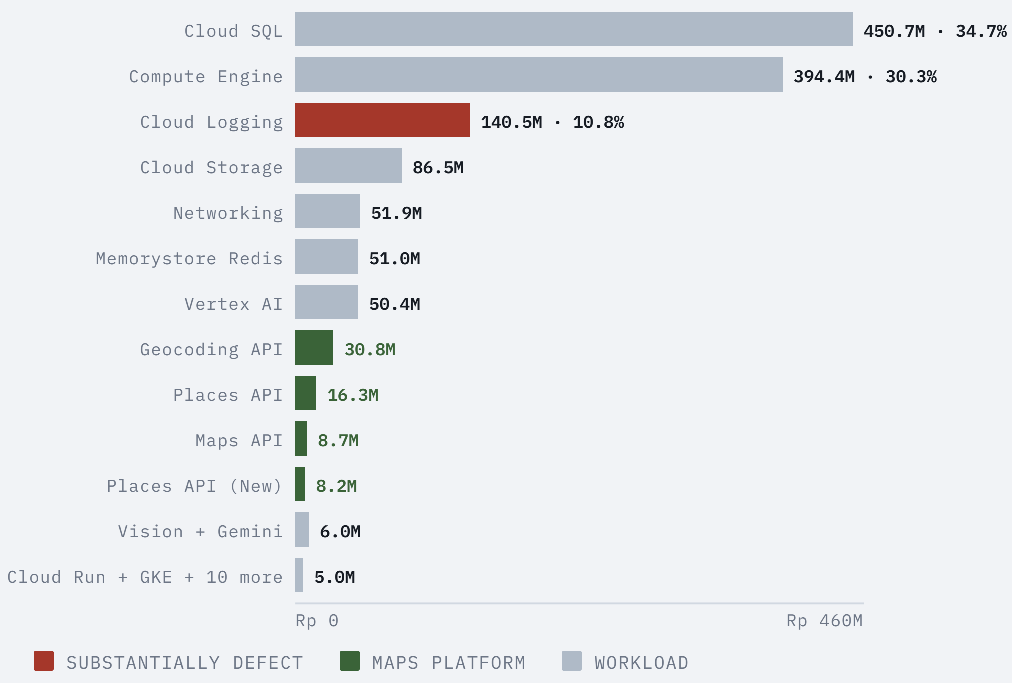
That number is arresting and it is the one most likely to be misread, so state it precisely: it is the bill for the platform that hosts most of BFI's origination *and* the ~26 shared `ms-*` data-plane services that **both platforms** depend on. **It is not the price of the workflow engine.**

TWO GROUPINGS WORTH NAMING, BECAUSE NEITHER APPEARS AS A LINE

Google Maps platform — Rp 63,979,414/month (Geocoding + Places + Maps + Places New). Larger than the entire LOS orchestration tier.

AI and OCR — Rp 56,357,315/month (Vertex AI + Vision + Gemini).

SERVICE BREAKDOWN, PRODUCTION PROJECT



The orchestration tier: ≈Rp58M

COMPONENT	AUGUST 2026, PROD	SOURCE
`prod-postgres-bpm-d2bpm` — vCPU + RAM	**Rp 40,930,777**	measured today, 30 daily line items
… storage, backup and remaining SKUs	≈ Rp 11,800,000	balance to the instance total
Cloud SQL instance, total	**≈ Rp 52,700,000**	**corroborated**
`ms-bpm` pods, production	≈ Rp 5,300,000	compare-architecture.md §8.4
Production LOS tier	**≈ Rp 58,000,000**	**—**
`ms-bpm` pods, SIT + UAT	≈ Rp 12,100,000	—
With non-prod pods	**≈ Rp 70,100,000**	**SIT/UAT/Sharia *databases* not separately pulled; true figure is higher**

THE CORROBORATION MATTERS

The Rp52.7M instance figure was previously derived once; an independent line-item pull today puts **vCPU + RAM alone at Rp40.93M across 30 days**, which is consistent and **rules out an order-of-magnitude error**. The BPM instance is 11.7% of the production project's Cloud SQL bill — a significant database, but one of many.

4.5%

THE TIER'S SHARE OF ITS OWN PRODUCTION PROJECT — AND 3.1% OF BRAVO'S TOTAL GCP SPEND

> **Whatever else is true, the workflow engine is not where Bravo's money goes.**

AGAINST LORA, LIKE FOR LIKE

	BRAVO LOS TIER	LORA TIER
What is in it	`ms-bpm` pods (Camunda + all human-task services) + `prod-postgres-bpm-d2bpm`	LPW/LTW workers, gateway, task service, schema service, Temporal Cloud contract, ArangoDB licence + GKE
Monthly, production	**≈ Rp 58M**	**≈ Rp 431M**
Applications, Aug 2026	76,446 (billing sheet) to ≈113k (engine meter)	171,479 (billing sheet) to ≈135k (Temporal meter)
Per application	**≈ Rp 510–760**	**≈ Rp 2,500–3,200**
Cost shape	~variable with pods, ~fixed on the database	~86% fixed: contracts and over-provisioned capacity

The denominators are the weakest part of both columns. The billing sheet's platform totals go 109.5k → 237.5k → 247.9k in two months, which nothing else supports, and the likeliest reading is that a LORA-originated application is *counted again* in Bravo when it books at go-live. If that is right, Bravo's denominator is inflated and its true per-application cost is higher than Rp510–760 — but not by enough to close a 4–5× gap.

Unit cost, and the direction it is moving

	JUN 2026	JUL 2026	AUG 2026	DIRECTION
Bravo applications	106,722	117,996	**76,446**	−35% in one month
Bravo support tickets (stable categories)	705	644	751	+82% since January
Bravo stuck-application rate	0.364%	0.348%	**0.699%**	worsening
LOS tier cost	~flat — one deployment, one database			flat

COST PER APPLICATION IS RISING WITH NO ENGINEERING CAUSE

Bravo's platform cost is essentially **fixed** and its volume is falling **by design**. Cost per application therefore rises for every application migrated away. Meanwhile the *ops* cost per application is rising faster: the stuck-application rate **doubled** between July and August on a shrinking book.

HELD LOOSELY: THE RESIDUAL-HARD-CASES CONFOUND

A draining platform keeps the products, branches and edge cases migrated last. That is a real confound and **it is not controlled for.** It is also consistent with a fixed-flowchart architecture under continuing product change.

NOT A FLOOR TO PLAN AGAINST

At **40k applications a month** the same ≈Rp58M is **≈Rp1,450 each**, and the gap to LORA *halves* without either team doing anything.

LORA's direction is the opposite, and this is the load-bearing fact for the migration case.

LORA's cost is ~86% fixed, so it absorbed **+23% Temporal-metered volume for +3% spend** between June and August. Its marginal cost per additional application is near zero. Bravo's is not zero — it is pods and database — but it is small.

Cost drivers ranked — and three that are defects

#	DRIVER	MONTHLY	NATURE
1	Cloud SQL, whole production project	Rp 450.7M	workload — the BPM instance is Rp52.7M of it
2	Compute Engine, whole production project	Rp 394.4M	workload
3	Cloud Logging	**Rp 140.5M**	**substantially defect**
4	Cloud Storage	Rp 86.5M	workload
5	**Non-production estate (`bravo-project-nonprod`)**	**Rp 573.5M**	**31% of all Bravo spend — governance**
6	**Google Maps platform**	**Rp 64.0M**	**workload, but see the geolocation defect**
7	Redis (Memorystore)	Rp 51.0M	workload — also the NIK-keyed cache flagged for PII
8	Vertex AI + Vision + Gemini	Rp 56.4M	workload

THE 404 STORM HAS A COST TAIL

`/bpm/v1/partnership-configuration/feature-configuration` returns 404 **266,767 times a week** and `PartnershipConfigurationController.getByApplicationId` is the **single busiest handler in the service** (73,056 spans in 48 hours). Every one of those is a servlet request, a JPA `repository.operation` against `prod-postgres-bpm-d2bpm`, and at least one log line. It is a small slice of a large database bill — but it is a slice being spent *to produce an error*, on two-thirds of surveyor sessions.

Cloud Logging at Rp140.5M is 2.4× the orchestration tier

`prod-ms-bpm` alone emits **1,116,135 log lines a week**, of which **525,181 (47%) are ERROR**. Two configuration choices explain most of that, and both are recorded in the code.

`LOGGERLEVEL: FULL` IS SET GLOBALLY ON FEIGN

...so **every one of the 113 clients logs complete request and response bodies**. That is PEFINDO, SLIK, Dukcapil and CONFINS payloads going into Cloud Logging as JSON — *a cost line and a data-protection exposure in the same breath.*

STACK TRACES ARE LOGGED ONE FRAME PER LINE

The top error “patterns” in the log stream are literally `at org.springframework.…` (874/week) and `at java.base/…`. **One exception becomes dozens of billable lines.** And one Camunda job failure produces *two* lines at *two* severities. Add **38,198 `ENGINE-09004` parse warnings a week** — pods re-parsing all 53 model files on every boot.

This is the largest single addressable line in Bravo’s bill, it is larger than the thing the migration would retire, and the fix is configuration.

Even a conservative 40% reduction is ≈Rp56M/month — **the whole LOS orchestration tier, recovered without migrating anything.**

Rp64M of Maps against failed geolocation — and 31% of spend nobody has opened

DEFECT 2 · RP64.0M OF MAPS AGAINST 20,690 GEOLOCATION FAILURES A WEEK

The Geocoding, Places and Maps APIs are **surveyor-platform costs**: a field surveyor's device resolving addresses and capturing position. Meanwhile RUM on the Surveyor Platform records `Error getting location: "[GeolocationPositionError]"` **10,365 times** and `Unable to get current position` **10,325 times** in seven days.

NOT CLAIMED AS CAUSAL

These two facts have not been connected before, and a browser geolocation failure and a server-side Maps call are *different layers*. But **Rp64M/month is being spent on location services on a console where location capture fails ~20,700 times a week**, and nobody has checked whether the failed attempts are also *billed* attempts. That is a day of work with a real number attached.

DEFECT 3 · NON-PRODUCTION IS 31% OF BRAVO'S SPEND

0.44

NON-PROD TO PROD RATIO — RP573.5M AGAINST RP1,300.3M

For comparison, roughly half of LORA's task-service and ArangoDB GKE lines are SIT/UAT — but LORA's non-prod sits inside a much smaller total.

NEVER EXAMINED

Nothing in this analysis establishes what is running in `bravo-project-nonprod` or whether it should be. At this size it is the *second-largest lever available* and it has never been looked at.

What the levers are worth

LEVER	WORTH / MONTH	AVAILABLE	CONFIDENCE
Cut Cloud Logging — disable global Feign body logging, log exceptions as one event, stop re-parsing warnings	**Rp 50–90M**	**now**	**high** — configuration only
Review `bravo-project-nonprod`	unknown, up to **Rp 573.5M**	**now**	**low** — never examined
Audit the Maps spend against failed geolocation	up to **Rp 64.0M**	**now**	**low** — needs a day of work first
Shorten Camunda history TTL from 90 d, or drop history level from `full`	part of Rp 52.7M	now, with a data-retention decision	medium — it is the audit trail after `application_status_log`
Fix the `feature-configuration` 404	small	now	**high** — removes 266,767 wasted DB round trips/week
Retire the Bravo LOS tier (the migration)	**Rp 58M prod, Rp 70–100M with non-prod**	**when the migration finishes**	**high**
For scale: the entire LORA Temporal Actions programme	*Rp 7.2M*	*now*	*high*

Two conclusions follow, and they are uncomfortable together.

The migration is still the single largest structural lever — and it is not the largest lever available this quarter.

Cloud Logging is bigger, available now, and depends on nothing. **A team that fixed logging configuration alone would save more in September than retiring the entire Bravo LOS tier will save when the migration completes.**

AND RETIRING BRAVO DOES NOT MAKE LORA'S TIER CHEAPER

LORA's ≈Rp431M is 86% fixed, and three of its four cost levers are **time-locked**: ~31% of node cost sits on a 3-year N2 CPU commitment; the Temporal commitment is renegotiable at **~March 2027**; and the five idle worker versions cannot retire while ~half of LORA's loans never reach a terminal state — a reliability defect presenting as a cost line. **The migration's cost case is *one platform instead of two, at near-zero marginal cost per application* — not *the cheaper platform wins.***

Eight actions

1 Cut the Cloud Logging bill · S, DO THIS FIRST — RP50–90M/MONTH

Turn off Feign `loggerLevel: full` globally and enable it per-client only where needed — this also removes upstream request bodies containing customer data from Cloud Logging. Log exceptions as a single structured event rather than one line per stack frame. Stop `ENGINE-09004` recurring by fixing the models.

The largest, fastest, lowest-risk saving in the document.

2 Open `bravo-project-nonprod` · S TO LOOK — UP TO RP573.5M/MONTH

31% of Bravo's spend has never been examined in this pack. Produce a service-level breakdown and a list of what is running and why.

Even a 20% reduction outbids the migration lever.

3 Check whether failed geolocation attempts are billed · S — UP TO RP64.0M/MONTH

Rp64M of Maps APIs on a console with 20,690 geolocation failures a week. Establish whether the two are related before deciding anything.

Owner identified 2026-09-10:

the call sites are in `bravo-surveyor-console` (`@react-google-maps/api` , 307k LOC), squad `LN` **— Team Surveyor & Verificator**

, which has 286 PR-gating tests to add a regression to once it is fixed.

4 Decide the Camunda history question explicitly · S–M

History level `full` with `P90D` on Cloud SQL is a real cost, and after day 91 the only remaining record of what happened to a loan is the trigger-written `application_status_log`. Either shorten the TTL and accept that, or keep it and stop calling it a cost problem — but

make it a decision rather than a default.

5 Fix the `feature-configuration` 404 · S

266,767 wasted round trips a week through the busiest handler in the service. Small money, real load, and a live defect regardless.

6 Get the billing owner to define the application counts · DAYS

The single largest source of error in the whole pack.

Every per-application figure here is a verified numerator over a disputed denominator. The `Bravo total app` and `Lora total app` columns need definitions — specifically on whether a LORA-originated application is counted again in Bravo at go-live.

7 Publish the LOS tier as a standing line · S

≈Rp58M is the number that matters for the migration decision, and it currently has to be reassembled by hand from line items each time. A saved FinOps view makes the migration's value trackable month to month.

8 Do not quote Rp1.6B as the migration saving · S, DOCUMENTATION

The Bravo estate is Rp1.887B and most of it is the shared data plane LORA depends on and which does not retire. The figure is

Rp58M prod, Rp70–100M with non-prod

. That claim was already withdrawn once; it should not come back.

30, 60, 90 days

Eight recommendations, phased. Most of this is **FinOps and Platform time, not Squad S&U engineering** — which matters, because the largest saving here is configuration, not migration.

SEQUENCING RULE

Money that is configuration comes before money that is migration. Cloud Logging at **Rp140.5M/month is 2.4× the entire ≈Rp58M tier the migration would retire**, and it is a config change available now. It starts in week 1. **The migration lever is not in this plan at all** — it lands when the migration finishes, and nothing here accelerates it.

SHARED WITH OTHER DOCUMENTS

Recommendation 5 is also **bravo-observability.md rec 1** and **bravo-delivery.md rec 1**; recommendation 3 pairs with **bravo-delivery.md rec 2**. Phased identically — do them once.

Days 0–30

START THE LARGEST SAVING, FIX THE DENOMINATORS

1 **Turn off global Feign `loggerLevel: full`**; re-enable per client only where genuinely needed. *S&U + Platform · 3 d.* **Largest single saving in the pack.** Upstream request bodies stop entering Cloud Logging — a cost line and a data-protection exposure closed together. Drop visible on the September bill.

6 **Define `Bravo total app` and `Lora total app`** — specifically whether a LORA-originated application is recounted in Bravo at go-live. *FinOps + billing owner · 2 d.*

5 **Fix the `feature-configuration` 404.** *S&U · 2 d* (investigation is in the observability plan).

7 **Publish the LOS tier as a standing FinOps view.** *FinOps · 1 d.* ≈Rp58M becomes a tracked chart.

2 **First look at `bravo-project-nonprod`** — service-level breakdown of Rp573.5M/month. *FinOps + Platform · 2 d.*

8 **Correct the Rp1.6B figure** wherever it survives in slides, decks and briefing notes. *EM · 0.5 d.*

Days 31–60

FINISH THE CUT, ANSWER THE OPEN QUESTIONS

1 **Log exceptions as one structured event**, not one line per stack frame; stop `ENGINE-09004` recurring by fixing the models. *S&U · 3 d.* Completes the **Rp50–90M/month** saving; ERROR share falls from 47% and the log stream becomes countable.

2 **`bravo-project-nonprod` reduction plan.** *FinOps + Platform · 3 d.* A decommissioning plan, or a written rationale for Rp573.5M/month.

3 **Audit Maps spend against failed geolocation.** *FinOps + S&U · 2 d.* Either a defect is fixed and money returns, or the spend is justified and stops being an open question.

4 **Decide the Camunda history question.** *S&U + Compliance · 2 d.* A recorded decision rather than a default.

Days 61–90

VERIFY THE SAVINGS LANDED

✓ No new cost work. **Phase 3 is verification, because a saving that is not measured on a subsequent invoice is a plan, not a saving.**

1 September and October **Cloud Logging** against the August baseline of Rp140.5M — target Rp50–90M/month reduction.

7 The **LOS tier standing view** across three months — confirm ≈Rp58M and its trend as volume drains.

6 **Per-application cost recomputed** on the agreed denominators — the first non-provisional unit cost in the pack.

Deferred

WITH TRIGGERS

4 **Camunda history implementation.** A data-retention change needing compliance sign-off, not an engineering task. *Trigger: after the phase-2 decision.*

2 **`bravo-project-nonprod` execution.** Phase 2 produces the plan; execution depends on what it finds. *Trigger: after the phase-2 plan.*

What changes when this is done

RECOMMENDATION	EXAMPLE WHEN DONE
Logging configuration fixed	Rp50–90M/month recovered — more than the migration lever — and PEFINDO/SLIK/Dukcapil/CONFINS request bodies stop landing in Cloud Logging.
Non-prod examined	The second-largest number in Bravo’s bill has an owner and a rationale, or a decommissioning plan.
Maps spend explained	Either a defect is fixed and Rp-tens-of-millions come back, or the spend is justified and stops being an open question.
History decision made	“What happened to this loan on day 120” has a documented answer instead of an accidental one.
Application counts defined	Every rate in this pack — cost per application, intervention rate, zero-intervention completion — stops being provisional.
LOS tier tracked	The migration’s financial value is a chart, not an argument.
Rp1.6B retired from the vocabulary	Nobody plans against a saving that was withdrawn in September 2026.

> The cheapest thing on this list is worth more than the migration, and it is a configuration change.